EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

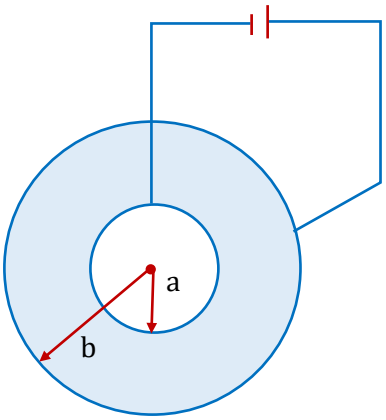
Microscopic Ohm's Law

1. An insulating pipe of cross-section area *'A'* contains an electrolyte which has two types of ions→ their charges being $-e$ and $+2e$. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v ($-ve$ ion) and $v/4$ ($+ve$ ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is

(A) $nevA/2$ (B) $nevA/4$ (C) $5\,nevA/2$ (D) $3\,nevA/2$

PCT047

2. A hollow spherical conductor having inner radius '*a*' and outer radius '*b*'. A constant potential difference is applied across inner and outer shell. Due to which current flows radially select the correct statement :-



(A) Drift velocity remains constant

(B) Drift velocity increases radially outward

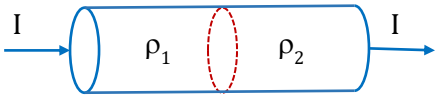
(C) Drift velocity decreases radially outward

(D) Drift velocity first decreases then increases radially outward

PCT050

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3. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-

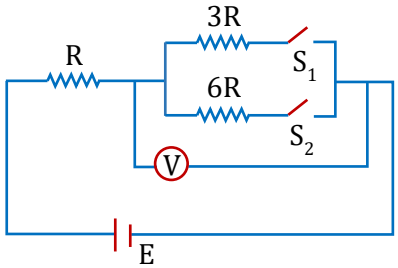


(A) Zero (B) $\frac{(\rho_1-\rho_2)I\varepsilon_0}{2}$ (C) $\varepsilon_0 I|\rho_1-\rho_2|$ (D) $\varepsilon_0 I|\rho_1+\rho_2|$

PCT052

Kirchoff's Laws

4. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then

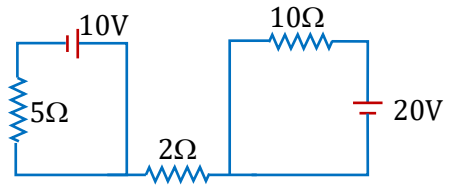


(A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$ (C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

PCT060

For More Join: @IITJEE_Advanced

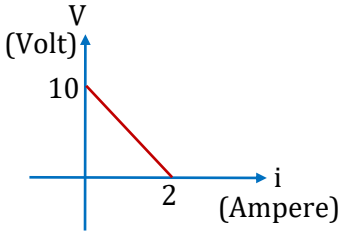
5. In the figure shown the current through 2Ω resistor is



(A) 2 *A* (B) 0 *A* (C) 4 *A* (D) 6 *A*

PCT065

6. A battery of emf *E* and internal resistance *r* is connected across a resistance *R*. Resistance *R* can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (*i*) passing through the resistance and potential difference (*V*) across it. Select the correct alternative.

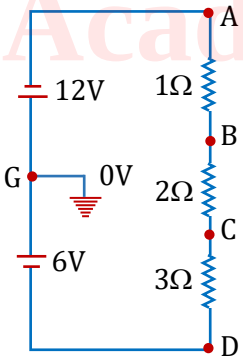


(A) internal resistance of battery is 5Ω
(B) emf of the battery is 20*V*
(C) maximum current which can be taken from the battery is 4*A*
(D) *V*-*i* graph can never be a straight line as shown in figure

PCT074

7. The potentials of points *A, B, C* and *D* as shown in figure are.

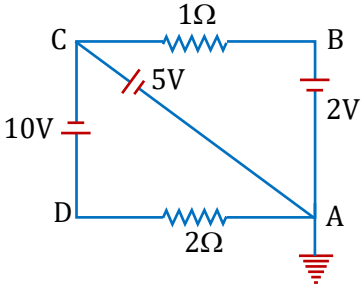
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(A) 12*V*, 11*V*, 9*V*, 6*V* (B) –11*V*, –12*V*, –9*V*, –6*V*
(C) –12*V*, –11*V*, –9*V*, –6*V* (D) –12*V*, –11*V*, –6*V*, –9*V*

PCT127

8. In the circuit shown in figure, the potential of *B* and the current through 2Ω resistance.



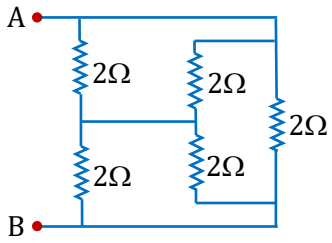
(A) 2 *V* and 7.5 *A* (B) 3 *V* and 7.5 *A* (C) 2 *V* and 5 *A* (D) 3 *V* and 5 *A*

PCT132

For More Join: @IITJEE_Advanced

Equivalent Resistance and Equivalent of Battery

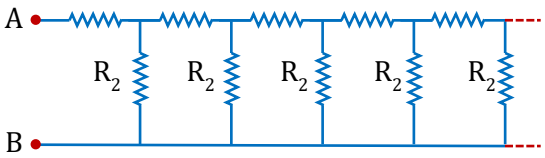
9. Find the equivalent resistance across AB



(A) 1 Ω (B) 2 Ω (C) 3 Ω (D) 4 Ω

PCT059

10. Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section.



(A) $R_1/R_2 = 1$ (B) $R_1/R_2 = 1/2$ (C) $R_1/R_2 = 2$ (D) $R_1/R_2 = 3$

PCT062

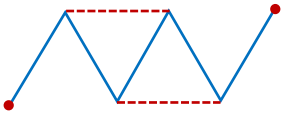
11. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are:-

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(A) 0.1 Ω, 4V (B) 0.2 Ω, 5V (C) 0.5 Ω, 10V (D) 0.7 Ω, 10V

PCT067

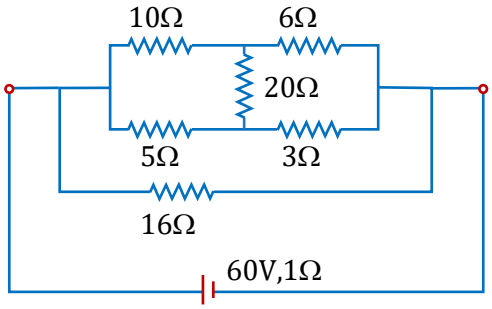
12. Five identical resistors each of resistance 1Ω are initially arranged as shown in the figure by clear lines. If two similar resistances are added as shown by the dashed lines, then change in resistance in final and initial arrangement is



(A) 2 Ω (B) 1 Ω (C) 3 Ω (D) 4 Ω

PCT096

13. Find equivalent resistance of circuit and current in 10Ω resistance?

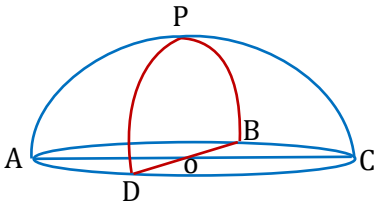


(A) 5Ω, 3A (B) 12Ω, 3A (C) 5Ω, 6A (D) 5Ω, 12A

PCT097

For More Join: @IITJEE_Advanced

14. A hemispherical network of radius a is made by using a conducting wire of resistance per unit length '*r*'. The equivalent resistance across *OP* will be.

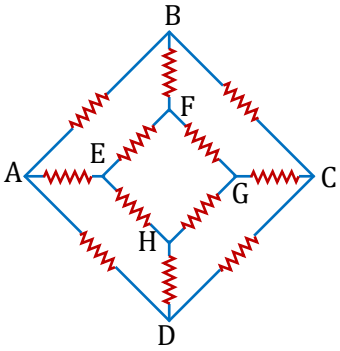


(A) $\left(\frac{2\pi+1}{3}\right)ar$ (B) $\left(\frac{\pi+3}{5}\right)ar$

(C) $\left(\frac{2\pi+3}{8}\right)ar$ (D) $\left(\frac{\pi+2}{8}\right)ar$

PCT123

15. In the network resistors each of value R shown in the fig., calculate the equivalent resistance between the junctions A and C :



(A) 12 R (B) $\frac{9R}{12}$ (C) $\frac{12R}{7}$ (D) $\frac{R}{12}$

PCT125

Heating Effect Due to Current and Electrical Instruments

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16. Two bulbs rated (25 W – 220 V) and (100 W – 220 V) are connected in series to a 440 V line. Which one is likely to fuse?

(A) 25 W bulb (B) 100 W bulb (C) Both bulbs (D) None

PCT083

17. Two bulbs one of 200 volts, 60 watts and the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

(A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

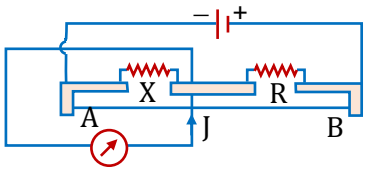
PCT085

18. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t. N number of similar cells is now connected in series with a wire of the same material and cross section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t. The value of N is:

(A) 4 (B) 6 (C) 8 (D) 9

PCT092

19. The figure shows a metre-bridge circuit, with AB = 100 cm, X = 12Ω and R = 18Ω, and the jockey J in the position of balance. If R is now made 8Ω, through what distance will J have to be moved to obtain balance?

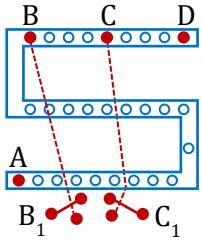


(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm

PCT100

For More Join: @IITJEE_Advanced

20. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between



(A) B and C (B) C and D (C) A and D (D) B_1 and C_1

PCT101

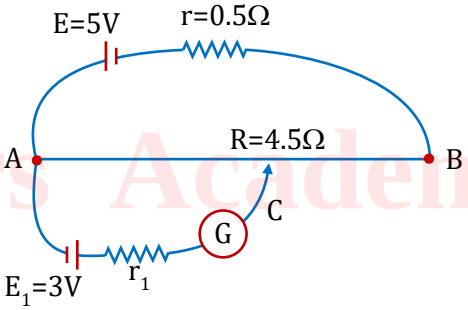
21. A galvanometer has a resistance of 20 Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(A) 0.01 Ω resistor connected across it (B) 0.02 Ω resistor connected across it

(C) 200 Ω resistor connected in series with it (D) 2000 Ω resistor connected in series with it

PCT102

22. In the given potentiometer circuit length of the wire AB is 3 m and resistance is R = 4.5 Ω. The length AC for no deflection in galvanometer is :-



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(A) 2 m (B) 1.8 m (C) Dependent on r_1 (D) None of these

PCT110

23. The balancing length for a 1.2 volts cell of 5 Ω internal resistance is 900 cm. If a resistance of 10 Ω is connected to the terminals of the cell, the potential difference and balancing length at the ends of resistance will be:

(A) 0.8 volt, 600 cm (B) 0.3 volt, 300 cm (C) 0.24 volt, 130 cm (D) 0.12 volt, 90 cm

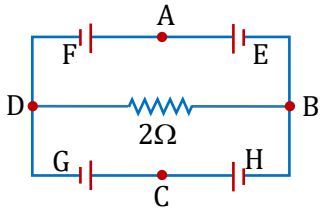
PCT118

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

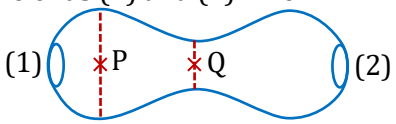
1. In the circuit shown *E, F, G* and *H* are cells of e.m.f. 2*V*, 1*V*, 3*V* and 1*V* respectively and their internal resistances are 2Ω, 1Ω, 3Ω and 1Ω respectively.



(A) $V_D - V_B = -2/13\ V$
(B) $V_D - V_B = 2/13\ V$
(C) $V_G = 21/13\ V$ = Potential difference across *G*
(D) $V_H = 19/13\ V$ = Potential difference across *H*

PCT071

2. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



(A) the current at the cross-section *P* equals the current at the cross-section *Q*
(B) the electric field intensity at *P* is less than that at *Q*.
(C) the rate of heat generated per unit time at *Q* is greater than that at *P*
(D) the number of electrons crossing per unit area of cross-section at *P* is less than that at *Q*.

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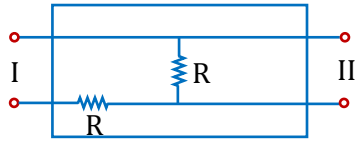
PCT091

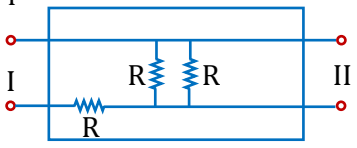
3. A micrometer has a resistance of 100 Ω and a full-scale range of 50 μ*A*. It can be used as a voltmeter, or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(*s*).

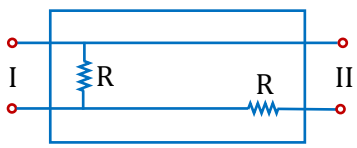
(A) 50 *V* range with 10 *k*Ω resistance in series.
(B) 10 *V* range with 200 *k*Ω resistance in series.
(C) 5 *mA* range with 1 Ω resistance in parallel.
(D) 10 *mA* range with 1 *k*Ω resistance in parallel.

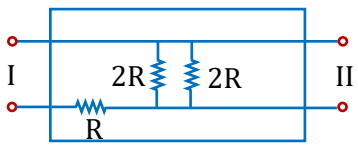
PCT103

4. If a battery of voltage *V* is connected across terminals I of the black box as shown in figure, an ideal voltmeter connected to terminals II gives a reading of *V*/2. If the same battery is connected to terminals II, an ideal voltmeter across terminals I reads *V*. Assuming that the black box consists only resistances, which of the following can be the possible combinations?

(A)


(B)


(C)


(D)


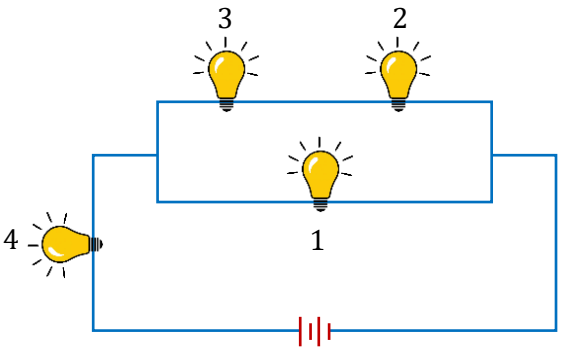
PCT133

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NOS. 5 TO 7

All bulbs consume same power. The resistance of bulb 1 is 36Ω.



5. What is the resistance of bulb 3 ?

(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

PCT088

6. What is the resistance of bulb 4 ?

(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

PCT089

7. What is the voltage output of the battery if the power of each bulb is $4W$?

(A) $12V$ (B) $16V$ (C) $24V$ (D) None of these

PCT090

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For More Join: @IITJEE_Advanced

EXERCISE (S)

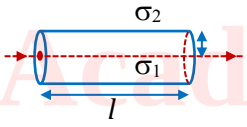
Microscopic Ohm's Law

1. (a) The current density across a cylindrical conductor of radius R varies according to the equation $J = J_0\left(1-\frac{r}{R}\right)$, where r is the distance from the axis. Thus, the current density is a maximum J_0 at the axis $r = 0$ and decreases linearly to zero at the surface $r = R$. Calculate the current in terms of J_0 and the conductor's cross-sectional area is $A = \pi R^2$.

(b) Suppose that instead the current density is a maximum J_0 at the surface and decreases linearly to zero at the axis so that $J = J_0\frac{r}{R}$. Calculate the current.

PCT048

2. A long conductor of circular cross-section has radius r and length l as shown in the figure. The conductivity of the material near the axis is σ_1 and increases linearly with the distance from axis and becomes σ_2 near the surface. Find the resistance of the conductor if the current enters from the one end and leaves from the other end.

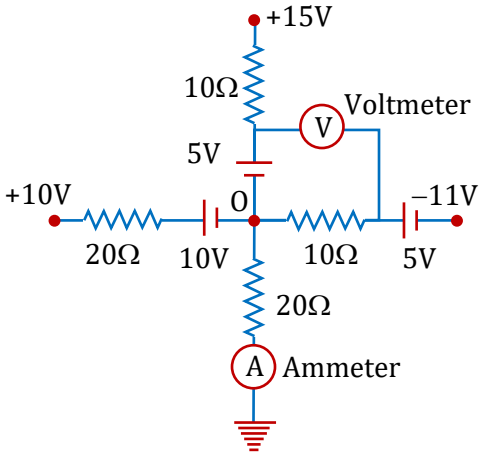


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PCT064

Kirchoff's Laws

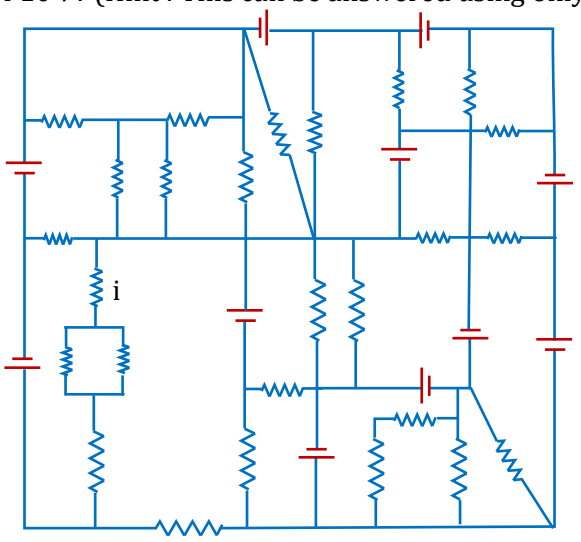
3. The potential of certain points in the circuit are maintained at the values indicated. The Voltmeter and Ammeter are ideal. Find the potential of the cross junction point in the circuit (at center O) and the readings of Voltmeter and Ammeter. All cells are ideal.



PCT128

For More Join: @IITJEE_Advanced

4. What is the magnitude and direction of current i in figure, where all resistance are 4 Ω and all battery have an emf of 10 V? (Hint : This can be answered using only mental calculation)

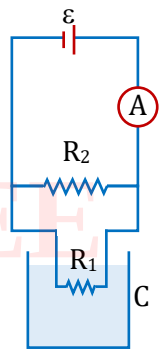


PCT134

Heating Effect Due to Current and Electrical Instruments

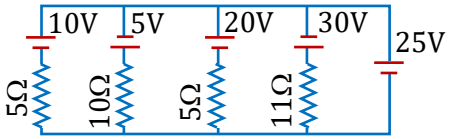
5. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature ($°C$) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flow through the coil and the ammeter shows 3A? The resistance $R_2 = 30\Omega$.

[Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = 4200 $J/kg°C$]



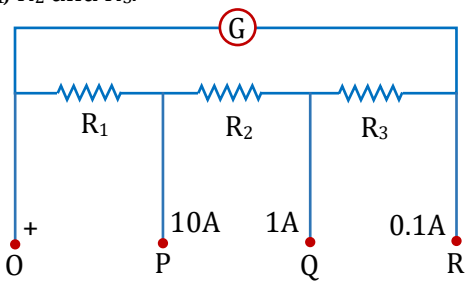
PCT093

6. Find the current through 25 V cell and power supplied by 20 V cell in the figure shown.



PCT095

7. The resistance of the galvanometer G in the circuit is 25Ω. The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0-10 A, if the terminals O and P are taken; range is 0–1 A between O and Q; range is 0–0.1 A between O and R. Calculate the resistance R_1, R_2 and R_3.



PCT105

For More Join: @IITJEE_Advanced

8. A galvanometer (coil resistance 99 Ω) is converted into a ammeter using a shunt of 1 Ω and connected as shown in the figure (i). The ammeter reads 3*A*. The same galvanometer is converted into a voltmeter by connected a resistance of 101 Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be 4/5 of the full-scale reading. Find

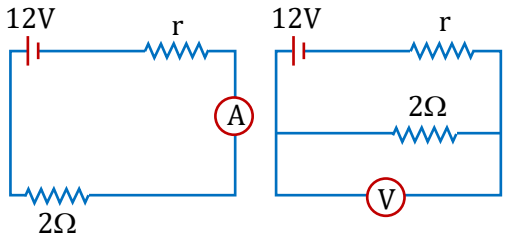


Fig. (i) **Fig. (ii)**

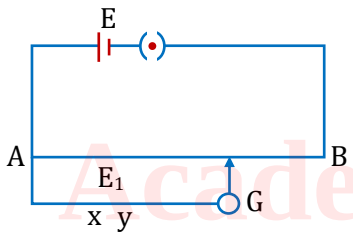
(i) internal resistance *r* of the cell

(ii) range of the ammeter and voltmeter

(iii) full scale deflection current of the galvanometer

PCT108

9. While doing an experiment with potentiometer it was found that the deflection is one sided and two cases are possible



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(i) the deflection decreased while moving from one end *A* of the wire to the end *B*;

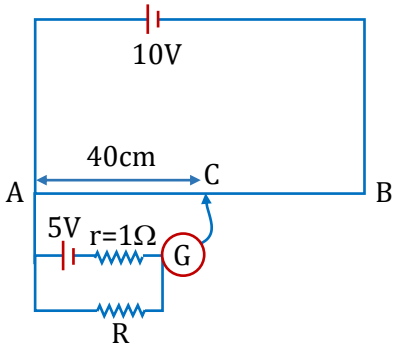
(ii) the deflection increased. while the jockey was moved towards the end *B*. Then

(a) Which terminal +ve or –ve of the cell E_1, is connected at *X* in case (i) and how is E_1 related to *E*?

(b) Which terminal of the cell E_1 is connected at *X* in case (ii)?

PCT113

10. A potentiometer wire *AB* is 100 *cm* long and has a total resistance of 10 ohm. If the galvanometer shows zero deflection at the position *C*, then find the value (in Ω) of unknown resistance *R*.

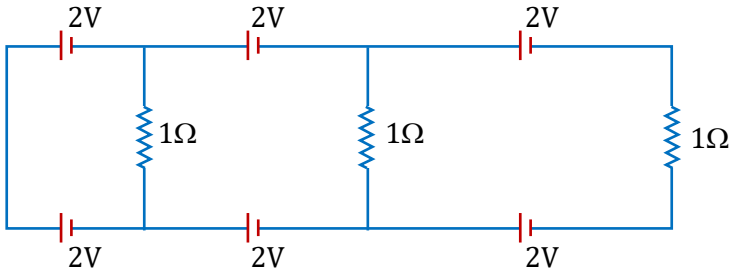


PCT114

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

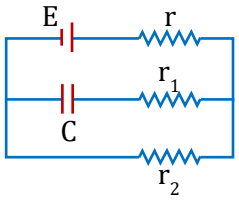
1. In the above circuit the current in each resistance is : **[JEE (Main) 2017]**



(1) 0 *A* (2) 1 *A* (3) 0.25 *A* (4) 0.5 *A*

PCT018

2. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance *C* will be : **[JEE (Main) 2017]**



(1) $CE\frac{r_2}{(r+r_2)}$ (2) $CE\frac{r_1}{(r_1+r)}$

(3) CE (4) $CE\frac{r_1}{(r_2+r)}$

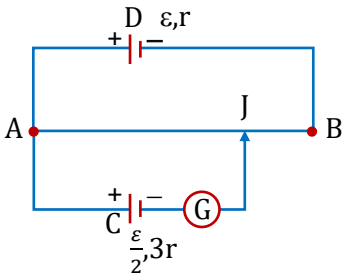
PCT019

3. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 *cm* of the potentiometer wire. If the cell is shunted by resistance of 5 Ω, a balance is found when the cell is connected across 40 *cm* of the wire. Find the internal resistance of the cell. **[JEE (Main) 2018]**

(1) 2 Ω (2) 2.5 Ω (3) 1 Ω (4) 1.5 Ω

PCT020

4. A potentiometer wire *AB* having length *L* and resistance 12 *r* is joined to a cell *D* of emf ε and internal resistance *r*. A cell *C* having emf ε/2 and internal resistance 3*r* is connected. The length *AJ* at which the galvanometer as shown in fig. shows no deflection is : **[JEE (Main) 2019]**

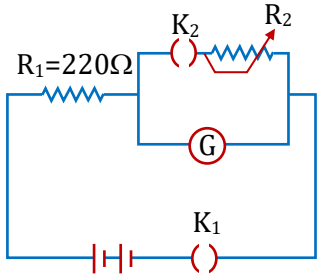


(1) $\frac{5}{12}L$ (2) $\frac{11}{24}L$

(3) $\frac{11}{12}L$ (4) $\frac{13}{24}L$

PCT022

5. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]: **[JEE (Main) 2019]**



(1) 12Ω (2) 25Ω

(3) 5Ω (4) 22Ω

PCT023

For More Join: @IITJEE_Advanced

6. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of $6mA$ it produces a deflection of 2°, its figure of merit is close to : **[JEE (Main) 2020]**

(1) $3 \times 10^{-3}\ A/div.$ (2) $333°\ A/div.$
(3) $6 \times 10^{-3}\ A/div.$ (4) $666°\ A/div.$

PCT024

7. A galvanometer of resistance G is converted into a voltmeter of range $0 - 1V$ by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2V$ will be : **[JEE (Main) 2020]**

(1) R_1 (2) $R_1 + G$ (3) $R_1 - G$ (4) G

PCT140

8. Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20 Ω. A voltmeter connected in the circuit measures 1.2 V. The internal resistance of each cell is **[JEE (Main) 2022]**

(1) 2.5 Ω (2) 4 Ω (3) 5 Ω (4) 10 Ω

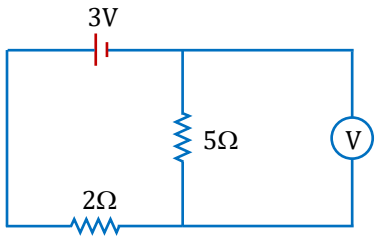
PCT025

9. A potentiometer wire of length 10 m and resistance 20 Ω is connected in series with a 25 V battery and an external resistance 30 Ω. A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____.

[JEE (Main) 2022]

PCT141

10. As shown in the figure, the voltmeter reads $2V$ across 5 Ω resistor. The resistance of the voltmeter is_____Ω. **[JEE (Main) 2023]**



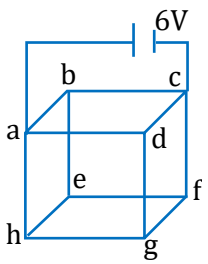
PCT130

11. The length of a metallic wire is increased by 20% and its area of cross section is reduced by 4%. The percentage change in resistance of the metallic wire is _____ . **[JEE (Main) 2023]**

PCT131

12. Twelve wires each having resistance 2Ω are joined to form a cube. A battery of 6 V emf is joined across point a and c. The voltage difference between e and f is____ V.

[JEE (Main) 2024]

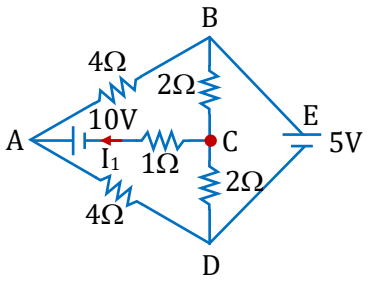


PCT142

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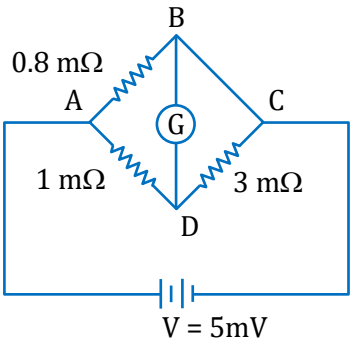
13. The current flowing through the 1 Ω resistor is $\frac{n}{10}$ A. The value of n is _____.

[JEE (Main) 2024]



PCT143

14. To measure the temperature coefficient of resistivity α of a semiconductor, an electrical arrangement shown in the figure is prepared. The arm BC is made up of the semiconductor. The experiment is being conducted at 25°C and resistance of the semiconductor arm is 3 $m\Omega$. Arm BC is cooled at a constant rate of 2°C/s. If the galvanometer G shows no deflection after 10s, then α is :



[JEE (Main) 2024]

(1) $-2 \times 10^{-2}\,°C^{-1}$ (2) $-1.5 \times 10^{-2}\,°C^{-1}$

(3) $-1 \times 10^{-2}\,°C^{-1}$ (4) $-2.5 \times 10^{-2}\,°C^{-1}$

PCT144

15. The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1s$ to $t = 2s$ is _____ C.

[JEE (Main) 2024]

PCT145

16. To measure the internal resistance of a battery, potentiometer is used. For R = 10 Ω, the balance point is observed at ℓ = 500 cm and for R = l Ω the balance point is observed at ℓ = 400 cm. The internal resistance of the battery is approximately :

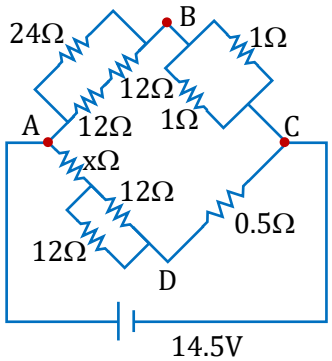
[JEE (Main) 2024]

(1) 0.2 Ω (2) 0.4 Ω (3) 0.1 Ω (4) 0.3 Ω

PCT146

17. The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is :

[JEE (Main) 2024]



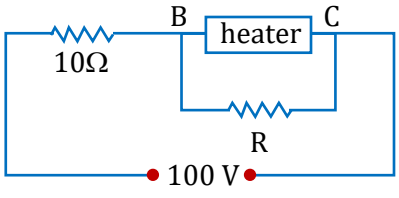
(1) 3 Ω (2) 9 Ω (3) 6 Ω (4) 42 Ω

PCT147

For More Join: @IITJEE_Advanced

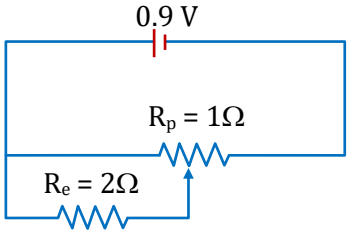
18. A heater is designed to operate with a power of 1000 *W* in a 100 *V* line. It is connected in combination with a resistance of 10 Ω and a resistance *R*, to a 100 *V* mains as shown in figure. For the heater to operate at 62.5 *W*, the value of *R* should be ..... Ω.

[JEE (Main) 2024]



PCT148

19.



Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance $R_P = 1\Omega$ as shown in the figure. An external resistance of $R_e = 2\Omega$ is connected via the sliding contact.

(1) 0.3 *A* (2) 1.35 *A* (3) 1.0 *A* (4) 0.9 *A*

[JEE (Main) 2025]

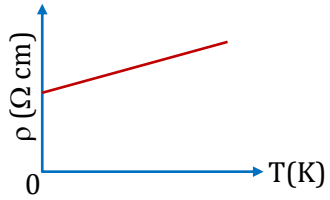
PCT202

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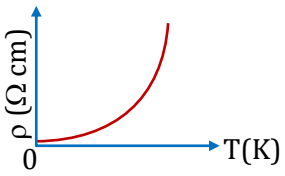
20. Which of the following resistivity (ρ) v/s temperature (*T*) curves is most suitable to be used in wire bound standard resistors?

[JEE (Main) 2025]

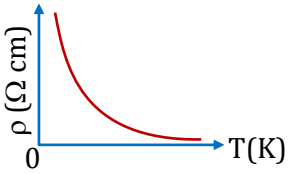
(1)



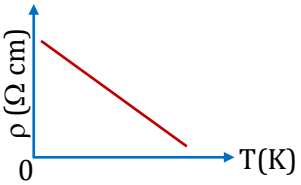
(2)



(3)



(4)

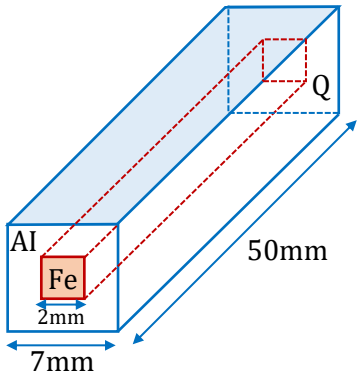


PCT203

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EXERCISE - JEE (Advanced) PYQ

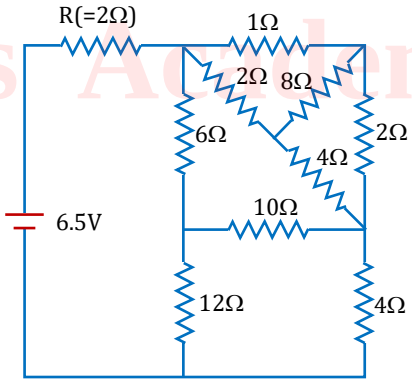
1. In an aluminium (Al) bar of square cross section, a square hole is drilled and is filled with iron (*Fe*) as shown in the figure. The electrical resistivities of *Al* and *Fe* are $2.7 \times 10^{-8}\ \Omega m$ and $1.0 \times 10^{-7}\ \Omega m$, respectively. The electrical resistance between the two faces *P* and *Q* of the composite bar is **[JEE (Advanced) 2015]**



(A) $\frac{2475}{64}\ \mu\Omega$ (B) $\frac{1875}{64}\ \mu\Omega$ (C) $\frac{1875}{49}\ \mu\Omega$ (D) $\frac{2475}{132}\ \mu\Omega$

PCT036

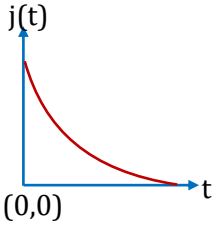
2. In the following circuit, the current through the resistor *R* (= 2Ω) is *I* Amperes. The value of *I* is : **[JEE (Advanced) 2015]**

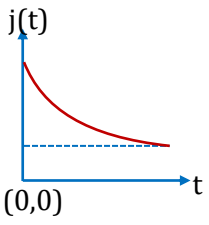


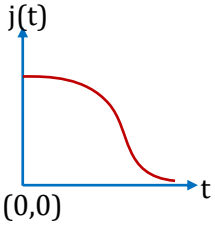
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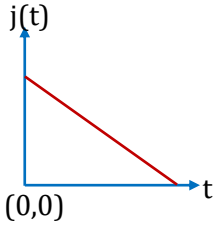
PCT037

3. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius *R*. At time $t = 0$, the space inside the cylinder is filled with a material of permittivity ε and electrical conductivity σ. The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density $j(t)$ at any point in the material? **[JEE (Advanced) 2016]**

(A) 

(B) 

(C) 

(D) 

PCT038

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4. Consider two identical galvanometers and two identical resistors with resistance R. If the internal resistance of the galvanometers $R_C < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true? **[JEE (Advanced) 2016]**

(A) The maximum voltage range is obtained when all the components are connected in series.

(B) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer.

(C) The maximum current range is obtained when all the components are connected in parallel.

(D) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel with both the resistors.

PCT039

5. Two identical moving coil galvanometer have 10 Ω resistance and full scale deflection at 2 μA current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with R = 1000 Ω resistor by using an ideal cell. Which of the following statement(s) is/are correct? **[JEE (Advanced) 2019]**

(A) The measured value of R will be $978\Omega < R < 982\Omega$.

(B) The resistance of the Voltmeter will be 100 kΩ.

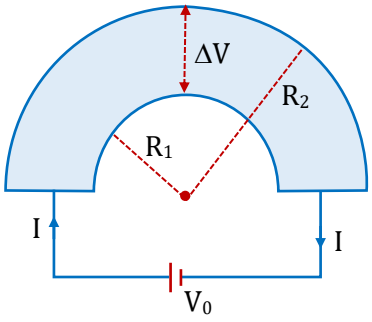
(C) The resistance of the Ammeter will be 0.02Ω (round off to 2nd decimal place)

(D) If the ideal cell is replaced by a cell having internal resistance of 5Ω then the measured value of R will be more than 1000 Ω.

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PCT040

6. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ. Its inner radius is R_1 and outer radius is R_2. If a voltage V_0 is applied between its two ends, a current I flows in it. In addition, it is observed that a transverse voltage ΔV develops between its inner and outer surfaces due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current). Then (figure is schematic and not drawn to scale) **[JEE (Advanced) 2020]**



(A) $I = \frac{v_0 t}{\pi\rho} In\left(\frac{R_2}{R_1}\right)$

(B) the outer surface is at a higher voltage than the inner surface

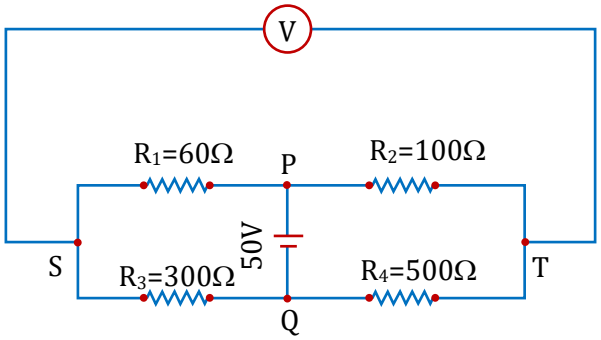
(C) the outer surface is at a lower voltage than the inner surface

(D) $\Delta V \propto I^2$

PCT041

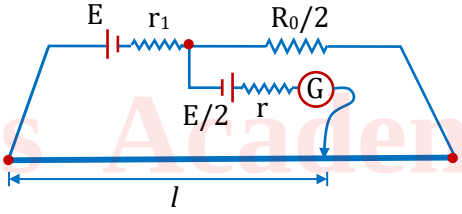
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7. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient $0.0004°C^{-1}$. If the temperature of R_3 is increased by 100°C, the voltage developed between S and T will be _____ volt. **[JEE (Advanced) 2020]**



PCT042

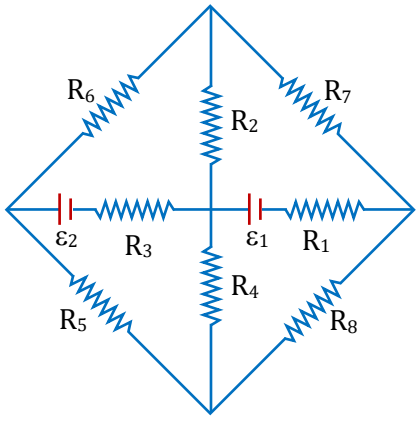
8. In order to measure the internal resistance r_1 of a cell of emf E, a meter bridge of wire resistance $R_0 = 50\ \Omega$, a resistance $R_0/2$, another cell of emf $E/2$ (internal resistance r) and a galvanometer G are used in a circuit, as shown in the figure. If the null point is found at $\ell = 72$ cm, then the value of r_1 = _____Ω. **[JEE (Advanced) 2021]**



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PCT043

9. The figure shows a circuit having eight resistances of 1 Ω each, labelled R_1 to R_8, and two ideal batteries with voltages $\varepsilon_1 = 12\ V$ and $\varepsilon_2 = 6\ V$. **[JEE (Advanced) 2022]**



Which of the following statement(s) is(are) correct?

(A) The magnitude of current flowing through R_1 is 7.2 A.

(B) The magnitude of current flowing through R_2 is 1.2 A.

(C) The magnitude of current flowing through R_3 is 4.8 A.

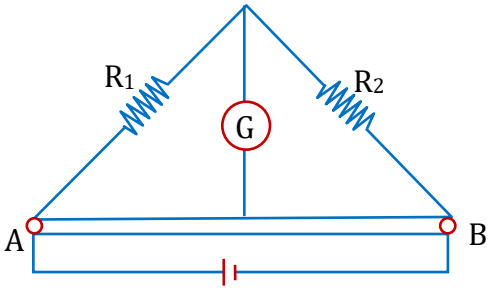
(D) The magnitude of current flowing through R_5 is 2.4 A.

PCT044

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10. Two resistances $R_1 = X\,\Omega$ and $R_2 = 1\,\Omega$ are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B. A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of X is _____.

[JEE (Advanced) 2022]

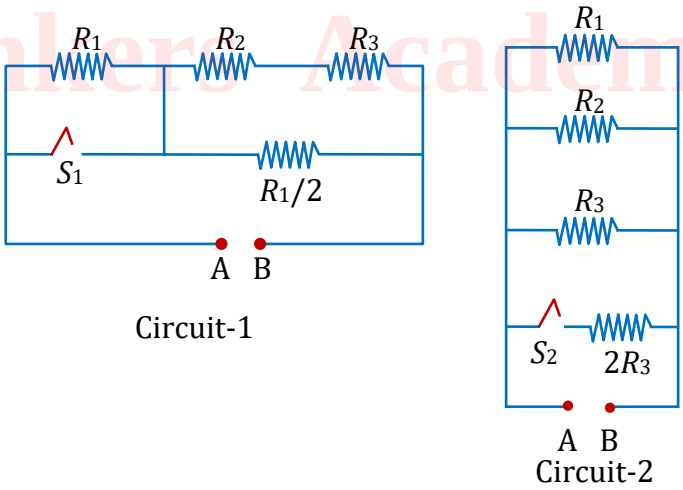


PCT045

11. In Circuit-1 and Circuit-2 shown in the figures, $R_1 = 1\,\Omega$, $R_2 = 2\,\Omega$ and $R_3 = 3\,\Omega$. P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively.

Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively.

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Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2022]**

(A) When a voltage source of 6 V is connected across A and B in both circuits, $P_1 < P_2$.

(B) When a constant current source of 2 Amp is connected across A and B in both circuits, $P_1 > P_2$.

(C) When a voltage source of 6 V is connected across A and B in Circuit-1, $Q_1 > P_1$.

(D) When a constant current source of 2 Amp is connected across A and B in both circuits, $Q_2 < Q_1$

PCT046

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	C	A	B	A	C	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	A	D	B	A	A	B	B	C
Que.	21	22	23							
Ans.	B	D	A							

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7
Ans.	ACD	ABCD	BC	AD	B	A	B

EXERCISE (S)

1. (a) $J_0A/3$; (b) $2J_0A/3$

2. $\frac{1}{R}=\frac{\pi r^2}{3l}(2\sigma_2+\sigma_1)$

3. $x=\frac{4}{3}V, 12\frac{1}{3}V, \frac{1}{15}A$

4. $4A$

5. 25

6. $12A, -20W$

7. $R_1 = 0.0278\ \Omega, R_2 = 0.25\ \Omega, R_3 = 2.5\ \Omega$

8. (i) 1.01Ω (ii) 0-5 A, 0-10V, (iii) 0.05 A

9. (a) $+ve, E_1 > E$ (b) $-ve$

10. 4

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	4	4	4	1	2	3	25	20
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	25	1	25	3	22	4	3	5	3	1

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	1	C	BC	AC	ACD	0.27	3	ABCD	5
Que.	11									
Ans.	ABC									

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Important Notes

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